Dabit Bista

PhD Candidate
Agricultural, Food, and Resource Economics
Michigan State University

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Research Interests

Applied microeconomics; agricultural and resource economics; behavioral and experimental economics; discrete choice experiments; risk preferences; production economics; conservation contracts; grazing systems.

Education

2023 - Present **Doctor of Philosophy, Agricultural and Food Economics**, *Michigan State University*, East Lansing, MI.
Dissertation: *Risk, Producer Decision-Making, and Production Performance in U.S. Cow-Calf Operations*.
Expected completion: May 2027.

2021 - 2023 **Master of Science, Agricultural Economics; Minor in Statistics**, *New Mexico State University*, Las Cruces, NM.
Thesis: *Consumer Willingness to Pay for New Chile Pepper with Improved Attributes*.

2015 - 2020 **Bachelor of Science in Agriculture**, *Agriculture and Forestry University*, Nepal.
Thesis: *Status and Economics of Honey Production in Jajarkot District of Nepal*.

Academic and Research Experience

Michigan State University, East Lansing, Michigan

2023 - Present **Graduate Research Assistant**, *Department of Agricultural, Food, and Resource Economics*.

Advisor: Dr. Frank Lupi.
Project: Metrics, Management, and Monitoring: An Investigation of Pasture and Rangeland Soil Health and Its Drivers (3M).

- Conduct applied economic research on U.S. cow-calf producer decisions, risk preferences, grazing management, and conservation participation.
- Design and analyze producer surveys and discrete choice experiments; clean and manage survey data; estimate econometric models and analyze heterogeneous responses.
- Study compensation for additional pasture rest in voluntary conservation contracts and links between experimentally elicited risk preferences and technical efficiency.
- Contribute to 3M research on pasture and rangeland soil health and grazing practices.
- Collect, clean, manage, and analyze financial records of 3M livestock producers, using tax documents, profit-and-loss statements, and cash-flow statements, to assess ranch costs, revenues, and financial performance.

New Mexico State University, Las Cruces, New Mexico

2021 - 2023 **Graduate Research Assistant**, *Department of Agricultural Economics and Agricultural Business*.

Advisor: Dr. Ram N. Acharya.
Project: Enhancing Pepper Production, Profitability, and Seed Yield in the Southwest U.S.

- Prepared enterprise crop budgets for different chile cultivars.

- Worked on input-output modeling of increased hot chile pepper production.
- Conducted economic profitability and marketing-window analysis.
- Conducted surveys of chile growers, and consumers.
- Analyzed consumer willingness to pay for pepper attributes.

Professional Experience

Prime Minister's Agriculture Modernization Project, Jajarkot, Nepal

Jan - Jun 2020 **Agriculture Intern**, *Bee/Citrus Zone*.
Supported agricultural research and extension, training facilitation, household surveys, profile-book preparation, and subsector business analysis. Contributed to beekeeping practice documentation, field monitoring, and evaluation.

Publications

Peer-Reviewed Journal Articles

2026 Stanley, P. L., Kuhl, A. S., Maciel, I. C., et al. **Merging rigor and relevance in grazingland research: A comprehensive social-ecological monitoring approach.** *Carbon Balance and Management*, 21, 67. DOI.

2024 Bista, D., Sapkota, S., Acharya, P., Acharya, R., & Ghimire, R. **Reducing energy and carbon footprint in semi-arid irrigated cropping systems through crop diversification.** *Heliyon*, 10(6).

2022 Bista, D., Sapkota, D., Paudel, H., & Adhikari, G. **Effect of foliar application of growth regulators on growth and yield of onion (*Allium cepa*).** *International Journal of Horticultural Science and Technology*, 9(2), 247-254.

2021 Sapkota, D., Adhikari, G., Regmi, R., & Bista, D. **Ecofriendly management of insect pests of cauliflower, *Brassica oleracea* var. *botrytis*, in Kapurkot, Salyan, Nepal.** *Biopesticides International*, 17(1).

2021 Adhikari, G., Bista, D., Bhattarai, A., & Paudel, H. **Effect of tuber size and nutrient source on the yield of yam (*Dioscorea* sp.).** *Acta Scientifica Malaysia*, 5(1), 17–19. DOI.

Work in Progress

Chapter 1 **Comparing Risk Preferences Elicited from Holt–Laury Lotteries and Discrete Choice Experiments.**
Investigates whether risk preferences elicited through lottery choices and conservation-contract choice experiments provide consistent measures for U.S. cow-calf producers. Estimates risk-aversion parameters from both methods and examines their correlation, heterogeneity, and implications for interpreting producer decisions under uncertainty.

Chapter 2 **The Price of Pasture Rest: Livestock Producers' Preferences for Conservation Contracts.**
Uses a discrete choice experiment to estimate producers' willingness to accept compensation for extending pasture rest under voluntary conservation contracts. Examines payment structures, grazing requirements, verification, and sponsor type, and models heterogeneous participation and potential acreage enrollment across producer groups and regions.

Chapter 3 **Risk Preferences and Technical Efficiency in U.S. Cow-Calf Operations.**
Investigates the relationship between experimentally elicited risk preferences and livestock production efficiency using stochastic frontier analysis. Estimates technical inefficiency while accounting for herd characteristics, grazing systems, production inputs, environmental conditions, and regional heterogeneity.

Conference and Research Presentations

2026 Conference paper: "Optimizing Grazing Management for Profit and Ecosystem Services: An Integrated Biogeochemical-Economic Model." Agricultural and Applied Economics Association Annual Meeting, Kansas City, MO, July 26-28.

2026 Oral presentation: "Economic Costs and Benefits of Grazing Strategies within the 3M Project." Society for Range Management Annual Meeting, Monterey, CA, February 7-11.

2023 Oral presentation: "Relative Risk and Profitability of Hand Picking and Mechanical Harvesting of Green Chilies." Research and Creativity Week, New Mexico State University (Third Place).

2022 Oral presentation: "Impact of Free Trade Agreements, Weather, and Product Branding on Chili Production in New Mexico." International Pepper Conference, August 26-28.

2022 Oral presentation: "Chili Production Risk and Profitability by Variety." International Pepper Conference, August 26-28.

2022 Poster presentation: "Impact of Rising Imports and Free Trade Agreements on New Mexico Chili Production." Open House Competition, New Mexico State University.

2022 Oral presentation: "Potential of Enhancing Profitability of Beekeeping in Nepal." SAEA Conference, New Orleans, LA, February 12-15.

2021 Oral presentation: "Emerging Challenges and Opportunities Faced by New Mexico Chile Industry." Research and Creativity Week, New Mexico State University.

Awards, Honors, and Scholarships

2024 Delisa Scholarship, Michigan State University.

2023 Third Place, Research and Creativity Week, New Mexico State University.

Fall 2021 Online and GA Assistantship, New Mexico State University.

Fall 2021 Grad Success Scholarship, New Mexico State University.

2015–2020 Merit Scholarship, Agriculture and Forestry University, Nepal.

Research and Technical Skills

Econometrics	Discrete choice models; mixed logit models; stochastic frontier analysis; willingness-to-pay analysis.
Research Methods	Experimental risk elicitation; survey design and administration; enterprise budgeting.
Data	Data cleaning, management, analysis, and visualization.
Software	Stata; R; SPSS; Microsoft Excel; Qualtrics; Ngene; JavaScript; basic SAS.

References

Dr. Frank Lupi
PhD Advisor and Research Supervisor
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